

ST314 Individual Project

Luqman Wong

Word Count: 5412

Question 1

1.1 Introduction

This study investigates three research questions using data from the 2022 Programme for International Student Assessment (PISA), which measures 15-year-olds' ability to use their reading, mathematics and science knowledge and skills to meet real-life challenges. The dataset contains 17,983 students from 747 schools in Canada, with variables collected from students, their parents, and their schools. The three research questions are: (i) the extent of school effects on students' mathematical ability, (ii) the individual-level and school-level predictors of students' mathematical ability, and whether the effect of any predictor differs for boys and girls, and (iii) variation in the effects of individual-level predictors across schools. The variables are described in Table 11. Data transformations are described in Appendix.

1.2 Descriptive Analysis

Univariate Statistics

Table 12 (Appendix) presents summary statistics for the continuous variables. **zmath** (maths score) has a mean of 0.061 and standard deviation of 0.996, consistent with standardisation (Figure 16). **age** has a very narrow range (15.33 to 16.42) meaning it's unlikely to be a strong predictor of **zmath** (Figure 17, Appendix). **hisced** (highest parental education) has a median of 8, corresponding to a Bachelor's degree (OECD, 2024). **homepos** is a scale derived from students' self-reported household possessions, used as a proxy for family wealth. The mean of 0.373 and wide range of -6.86 to 5.63 indicates considerable variation in family wealth across the sample. **stubeha** (student behaviour) has a mean of 0.566, with a right-skewed distribution (median = 0.74).

Table 13 presents the frequencies of the categorical variables. The sample is well balanced by gender with a near 50:50 split. The majority of students (74.0%) are native-born, with 2nd generation and 1st generation immigrants comprising 12.4% and 13.6% respectively. English is the most common language spoken at home (69.1%), followed by French (16.9%) and other languages (14.0%). The school sample is predominantly government schools (92.3%), with only 53 private schools (7.7%).

Bivariate Associations

Individual-Level Predictors

Figures 1 and 2 display the distribution of **zmath** by gender and immigration background respectively. Males have a slightly higher median maths score than females, suggesting a

modest gender gap in mathematical ability. For immigration background, native students have the lowest median maths score, followed by 1st generation immigrants and 2nd generation performing the best.

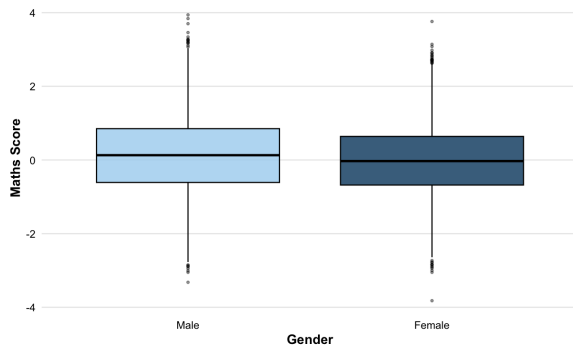


Figure 1: `zmath` by `female`

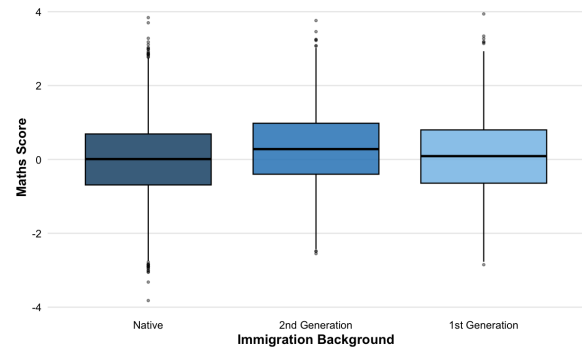


Figure 2: `zmath` by `immig`

Figure 3 shows that French speakers have a notably higher median maths score than English or other language speakers. Figures 4 and 5 show positive associations between `zmath` and both continuous individual-level predictors `hisced` and `homepos`. Both are consistent with the expectation that students from wealthier or more educated family backgrounds perform better in mathematics.

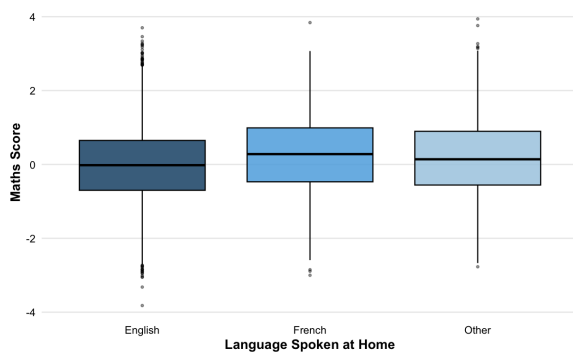


Figure 3: `zmath` by `langn`

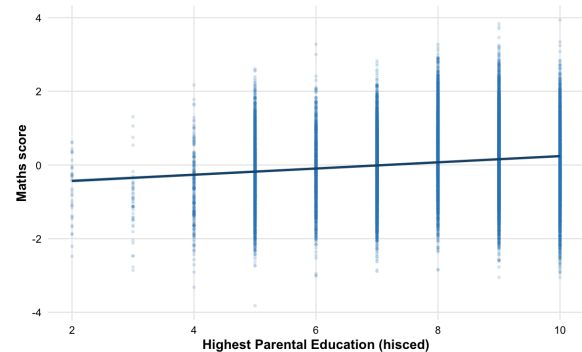


Figure 4: `zmath` vs `hisced`

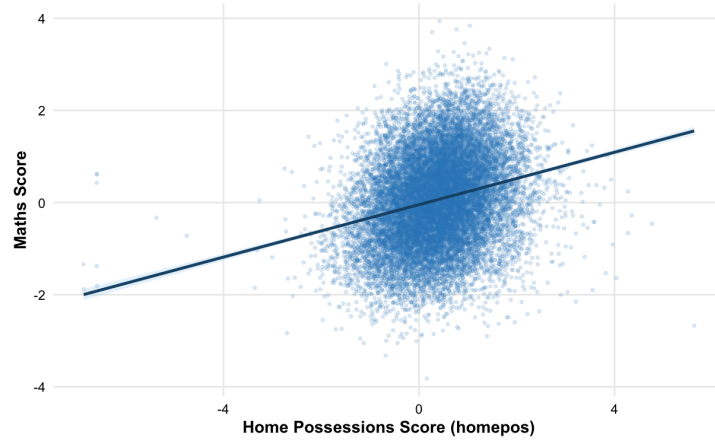


Figure 5: `zmath` vs `homepos`

School-Level Predictors

Figure 6 shows a clear difference in school median maths scores by school type, with private schools having a substantially higher median than government schools. Figure 7 shows a clear negative association between `stubeha` and mean school maths scores.

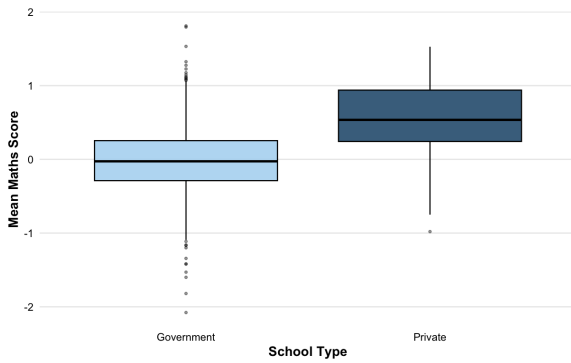


Figure 6: `zmath` by `schprivate`

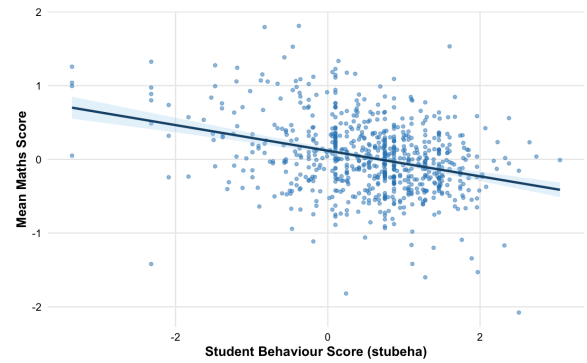


Figure 7: `zmath` vs `stubeha`

Between- and Within-School Variation

To explore the extent of school effects on `zmath`, a variance components model was fitted with `schoolid` as the only random effect and no predictors:

$$\text{zmath}_{ij} = \beta_0 + u_j + e_{ij} \quad (1)$$

where $u_j \sim N(0, \sigma_u^2)$ is the school-level random effect and $e_{ij} \sim N(0, \sigma_e^2)$ is the individual-level residual, for student i in school j . The results are presented in Table 1.

Table 1: Variance Components Model

Intraclass Correlation	Adjusted ICC:	0.170
Random Effects	Between-School Variance:	0.169
	Within-School Variance:	0.824
Fixed Effects	Estimate:	0.035
	Standard Error:	0.017

The estimated between-school variance is 0.169 and the within-school variance is 0.824. The intraclass correlation coefficient (ICC), calculated as $\hat{\sigma}_u^2 / (\hat{\sigma}_u^2 + \hat{\sigma}_e^2)$, is 0.170, indicating that 17.0% of the total variation in maths scores is attributable to differences between schools, with the remaining 83.0% due to variation among students within schools. This suggests there are meaningful school effects on mathematical ability, justifying the use of multilevel modelling.

To formally test the significance of school effects, a likelihood ratio test (LRT) was carried out comparing the variance components model against an ordinary least squares (OLS) model with no school effect. The hypotheses are:

$$H_0 : \sigma_u^2 = 0$$

$$H_1 : \sigma_u^2 > 0$$

Since σ_u^2 is constrained to be non-negative, the standard p-value is halved to account for the one-sided alternative. The LRT statistic is 2071.553 on 1 degree of freedom, giving a one-sided p-value < 0.001 . We therefore reject H_0 and conclude that there are statistically significant differences between schools in average maths scores, confirming that school membership should be treated as a random effect.

1.3 Model Selection

The Model selection procedure consists of six steps: beginning with a null model, adding individual-level predictors, testing gender interactions, adding school-level predictors, testing random slopes, and finally testing cross-level interactions. At each step, predictors were retained only if they significantly improved upon the model from the previous step at the 5% significance level. This was assessed by comparing the log-likelihoods of the two nested models via an LRT.

Step 1: Null Model

The variance components model fitted in Section 1.2 serves as the baseline

Step 2: Individual-Level Predictors

All individual-level predictors were added simultaneously:

$$\begin{aligned} \text{zmath}_{ij} = & \beta_0 + \beta_1(\text{Female})_{ij} + \beta_2(\text{2nd Gen})_{ij} + \beta_3(\text{1st Gen})_{ij} \\ & + \beta_4(\text{French})_{ij} + \beta_5(\text{Other})_{ij} + \beta_6(\text{hiscd.c})_{ij} \\ & + \beta_7(\text{homepos})_{ij} + \beta_8(\text{age.c})_{ij} + u_j + e_{ij} \end{aligned} \quad (2)$$

All predictors had $|t| > 1.96$, indicating individual significance at the 5% level. Backward elimination produced no removals, so all individual variables are retained. The results are summarised in Table 14 (Appendix).

Step 3: Gender Interactions Among Individual-Level Predictors

To address research question (ii), two-way interactions between **female** and each individual-level predictor were tested one at a time against Model 2. The null and alternative hypotheses for each test take the form:

$$\begin{aligned} H_0 : \beta_{\text{interaction}} &= 0 \\ H_1 : \beta_{\text{interaction}} &\neq 0 \end{aligned}$$

The results are presented in Table 15. Only the **female** $\times$ **hiscd.c** interaction was statistically significant at the 5% level. Model 3 adds $\beta_9(\text{Female} \times \text{hiscd.c})_{ij}$ to Model 2.

Step 4: School-Level Predictors

The two school-level predictors, **schprivate** and **stubeha**, were added to Model 3 using forward selection. Each was tested one at a time using an LRT of the form:

$$\begin{aligned} H_0 : \beta_{\text{school}} &= 0 \\ H_1 : \beta_{\text{school}} &\neq 0 \end{aligned}$$

`schprivate` was added first ($\chi^2 = 71.62$, $df = 1$, $p < 0.001$), followed by `stubeha` ($\chi^2 = 37.96$, $df = 1$, $p < 0.001$). Both school-level predictors were highly significant and were retained. The resulting model (Model 4) is shown in the appendix.

Step 5: Random Slopes

To address research question (iii), random slopes for each individual-level predictor were tested one at a time against Model 4. Since the random slope variance σ_{u1}^2 is constrained to be non-negative, a mixture chi-squared test was used, comparing the LRT statistic against $0.5\chi_1^2 + 0.5\chi_2^2$. The hypotheses are:

$$H_0 : \sigma_{u1}^2 = 0$$

$$H_1 : \sigma_{u1}^2 > 0$$

The results are presented in Table 16 (Appendix). Random slopes for `immig` and `langn` failed to converge despite using the Nelder-Mead optimiser and were therefore abandoned. Of the remaining four predictors, random slopes for `female`, `hisced.c`, `homepos` and `age.c` were all statistically significant at the 5% level.

To determine the best combination of random slopes, pairwise combinations of the three strongest candidates were tested, as shown in Table 2.

Table 2: LRT Results for Combined Random Slope Models

Random Slopes	LR Statistic	<i>p</i> -value	Converged?
<code>female + hisced.c</code>	69.879	< 0.001***	Yes
<code>female + homepos</code>	48.786	< 0.001***	Yes
<code>hisced.c + homepos</code>	55.911	< 0.001***	Yes

*** significant at 0.1%

The combination of `female` and `hisced.c` yielded the largest LRT statistic (LR = 69.88, $p < 0.001$). Model 5 therefore includes random slopes for both `female` and `hisced.c`:

$$\begin{aligned}
\text{zmath}_{ij} = & \beta_0 + \beta_1(\text{Female})_{ij} + \beta_2(\text{2nd Gen})_{ij} + \beta_3(\text{1st Gen})_{ij} \\
& + \beta_4(\text{French})_{ij} + \beta_5(\text{Other})_{ij} + \beta_6(\text{hisced.c})_{ij} \\
& + \beta_7(\text{homepos})_{ij} + \beta_8(\text{age.c})_{ij} + \beta_9(\text{Female} \times \text{hisced.c})_{ij} \\
& + \beta_{10}(\text{schprivate})_j + \beta_{11}(\text{stubeha})_j \\
& + (u_{0j} + u_{1j} \text{Female}_{ij} + u_{2j} \text{hisced.c}_{ij}) + e_{ij}
\end{aligned} \tag{5}$$

where u_{0j} , u_{1j} , u_{2j} are school-level random effects and $e_{ij} \sim N(0, \sigma_e^2)$.

Step 6: Cross-Level Interactions

To investigate whether the effects of individual-level predictors with random slopes vary systematically with school-level variables, four cross-level interactions between each random slope (**female**, **hisced.c**) and each school-level predictor (**schprivate**, **stubeha**) were tested against Model 5. The hypotheses for each take the form $H_0 : \beta_{\text{cross}} = 0$ vs $H_1 : \beta_{\text{cross}} \neq 0$. The results are presented in Table 17 (Appendix).

The interaction **hisced.c** $\times$ **schprivate** failed to converge despite rescaling and was abandoned. The remaining three interactions were not statistically significant at the 5% level, and hence not retained.

1.4 Model Interpretation

Fixed Effects

Table 18 presents the fixed effects estimates for the final model (Model 5). Since **zmath** is standardised, all estimates are interpreted in standard deviation units, with each interpretation given holding all other variables constant.

Intercept: The intercept ($\hat{\beta}_0 = 0.003$, $p = 0.883$) represents the expected **zmath** at the reference level (a male, native, English-speaking student attending a government school with mean age, mean parental education, and **homepos** = 0). It is not statistically significant.

Gender (female) and Parent education (hisced.c): Due to the significant interaction between **female** and **hisced.c**, the effects of gender and parental education will be considered jointly. Because **hisced** has been mean-centred, the main effect of **female** ($\hat{\beta}_1 = -0.134$, $p < 0.001$) now has a direct and meaningful interpretation: at the mean level of parental education, females score 0.134 standard deviations lower than males in mathematics. The interaction coefficient ($\hat{\beta}_9 = 0.027$, $p = 0.001$) indicates that the gender gap narrows by 0.027 standard deviations for each one-unit increase in **hisced**.

For example, at the maximum parental education level (`hisced` = 10, corresponding to `hisced.c` $\approx$ 2):

$$\hat{\beta}_1 + \hat{\beta}_9 \times 2 = -0.134 + 0.027 \times 2 = -0.080$$

This suggests that higher parental education is particularly beneficial for girls' math scores relative to boys.

Furthermore, viewing the interaction from the perspective of parental education, the main effect of `hisced.c` ($\hat{\beta}_6 = 0.009$, $p = 0.156$) represents the effect of parental education for male students at mean parental education, which is not statistically significant. For female students, the total effect is $\hat{\beta}_6 + \hat{\beta}_9 = 0.009 + 0.027 = 0.036$ per unit increase in `hisced.c`, which confirms that parental education is more beneficial for girls' maths scores. This is consistent with the OECD research which confirms that boys typically outperform girls in mathematics, partly due to gender stereotypes and biases from parents and teachers (OECD, 2024). The results suggest that educated parents may hold less stereotypical beliefs and invest more equally in their children, helping to narrow the gender gap.

Immigration background: Relative to native students, 2nd generation immigrants score 0.107 standard deviations higher ($p < 0.001$). 1st generation immigrants score 0.051 standard deviations above natives, though this effect is marginally insignificant at the 5% level ($p = 0.052$). This is consistent with the “immigrant paradox” whereby children of immigrants outperform native-born peers. Kao and Tienda (1995) attribute this to parents holding a belief that education is the primary route to social mobility in their new country.

Language spoken at home: French speakers score 0.129 standard deviations higher than English speakers ($p < 0.001$), even after controlling for immigration background and other predictors. This effect likely reflects the influence of Quebec's French-language school system, which is known for its strong mathematics curriculum and consistently performs above the Canadian average (CMEC, 2022). Students speaking other languages at home score 0.057 standard deviations above English speakers ($p = 0.030$), a smaller but still significant effect.

Home possessions: `homepos` has the strongest individual-level fixed effect ($\hat{\beta}_7 = 0.242$, $p < 0.001$). Each one-unit increase in the home possessions score is associated with a 0.242 standard deviation increase in `zmath`. Multiplying by the standard deviation of `homepos` gives $0.242 \times 0.844 = 0.204$ standard deviations increase in `zmath`.

Age: The mean-centred age coefficient ($\hat{\beta}_8 = 0.108$, $p < 0.001$) indicates that a one-year increase in age is associated with a 0.108 standard deviation increase in maths scores. While statistically significant, the effect is very small in practical terms given the narrow age range in the data.

School type: Students in private schools score 0.345 standard deviations higher than those in government schools ($p < 0.001$). This likely reflects the greater resources, more selective student intake, and potentially higher quality teaching in private schools.

Student behaviour: `stubeha` has a significant negative fixed effect ($\hat{\beta}_{11} = -0.098$, $p < 0.001$). Each one unit increase in the student behaviour score is associated with a 0.098 standard deviation decrease in maths scores. It is worth noting that the direction of causality is not clear. While disruptive student behaviour may hinder learning and reduce maths performance, it is equally plausible that students who struggle academically are more likely to exhibit disruptive behaviour.

Random Effects

The final model (Model 5) includes a random intercept and random slopes for `female` and `hisced.c`, with estimated variances and covariances presented in Table 19.

Random Intercept Variance

The random intercept variance $\hat{\sigma}_{u0}^2 = 0.129$ represents the between-school variance in average maths scores for the reference level. The 95% plausible range for a school's intercept is:

$$\begin{aligned}\hat{\beta}_0 \pm 1.96\sqrt{\hat{\sigma}_{u0}^2} &= 0.003 \pm 1.96\sqrt{0.1287} \\ &= (0.706, -0.700)\end{aligned}$$

Therefore, across schools, we expect 95% of baseline average maths scores to lie in the range $(-0.700, 0.706)$, a wide interval reflecting substantial between-school heterogeneity even after accounting for all fixed effects.

The random slope variance for `female` $\hat{\sigma}_{u1}^2 = 0.0117$ represents the between-school variance in the gender gap. The 95% plausible range for the school-specific gender gap is:

$$\begin{aligned}\hat{\beta}_1 \pm 1.96\sqrt{\hat{\sigma}_{u1}^2} &= -0.134 \pm 1.96\sqrt{0.0117} \\ &= (-0.346, 0.078)\end{aligned}$$

Therefore, across schools, we expect 95% of school-specific gender gaps to lie in the range $(-0.346, 0.078)$. This interval indicates that while girls underperform boys in the majority of schools, some schools show the reverse of this.

The random slope variance for `hisced.c` $\hat{\sigma}_{u2}^2 = 0.0030$ represents the between-school variance in the effect of parental education on maths scores, evaluated for male students. The 95% plausible range for the school-specific `hisced.c` slope (males) is:

$$\hat{\beta}_6 \pm 1.96\sqrt{\hat{\sigma}_{u2}^2} = 0.009 \pm 1.96\sqrt{0.0030}$$

$$= (-0.099, 0.116)$$

Therefore, across schools, the effect of parental education for male students is expected to lie in the range $(-0.099, 0.116)$, again spanning both directions and indicating meaningful school-to-school variation in how strongly parental education is associated with maths performance.

Covariances and Correlations Between Random Effects

The covariance between the random intercept and the female slope is $\hat{\sigma}_{u01} = -0.030$, giving a correlation of $\hat{\rho}_{u01} = -0.787$. Since $\hat{\beta}_1 < 0$ (the gender gap is negative, i.e. girls score lower), this strong negative correlation indicates girls are relatively more disadvantaged in high-performing schools. This is a fanning-out effect whereby this gender gap in math scores widens as school quality improves.

The covariance between the random intercept and the `hisced.c` slope is $\hat{\sigma}_{u02} = 0.010$, giving a correlation of $\hat{\rho}_{u02} = 0.504$. This moderate positive correlation indicates that in schools with higher average maths scores, the effect of parental education tends to be stronger.

The covariance between the two random slopes is $\hat{\sigma}_{u12} = -0.004$, giving a correlation of $\hat{\rho}_{u12} = -0.739$. This strong negative correlation indicates that in schools where the effect of parental education is larger, the gender gap tends to be smaller (or more favourable to girls). This is consistent with the fixed-effect finding that parental education disproportionately benefits girls.

Between-School Variance

Since `female` and `hisced.c` both have random slopes, we examine each in turn.

For `female` the between-school variance differs for male and female students. Using the formula:

$$\text{Var}(u_{0j} + u_{1j} \text{female}_{ij}) = \hat{\sigma}_{u0}^2 + 2\hat{\sigma}_{u01}(\text{female}) + \hat{\sigma}_{u1}^2(\text{female})^2$$

and

$$\text{ICC} = \frac{\hat{\sigma}_{u0}^2 + 2\hat{\sigma}_{u01}(\text{female}) + \hat{\sigma}_{u1}^2(\text{female})^2}{\hat{\sigma}_{u0}^2 + 2\hat{\sigma}_{u01}(\text{female}) + \hat{\sigma}_{u1}^2(\text{female})^2 + \hat{\sigma}_e^2}$$

evaluating at male ($\text{female} = 0$) and female ($\text{female} = 1$):

Table 3: Between-School Variance and ICC by Gender

Gender	Between-School Variance	ICC
Male	0.1287	0.143
Female	0.079	0.093

The between-school variance is considerably larger for males (0.129) than for females (0.079), reflected in a higher ICC for males (0.143 vs. 0.093). This means that for boys, a greater share of the variation in maths scores is attributable to which school they attend, whereas girls' scores are relatively less influenced by school context.

As `hisced.c` also has a random slope, the between-school variance additionally depends on parental education. Using the formula:

$$\hat{\sigma}_{u0}^2 + 2\hat{\sigma}_{u02}(\text{hisced.c}) + \hat{\sigma}_{u2}^2(\text{hisced.c})^2$$

Figure 8 shows how the between-school variance changes across the observed range of `hisced.c`. The variance increases with parental education, reflecting the positive intercept–slope correlation ($\hat{\rho}_{u02} = 0.504$): schools differ more in their maths outcomes among students from highly educated families than among those from less educated families.

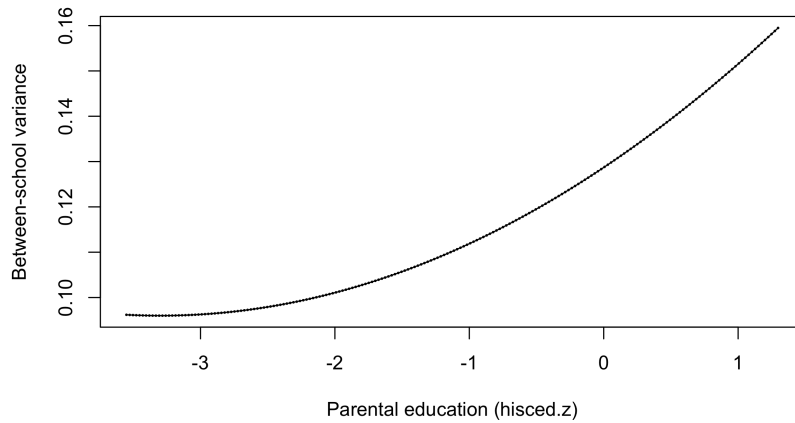


Figure 8: Between-school variance as a function of parental education (`hisced.c`)

1.5 Limitations

Several limitations of this analysis should be acknowledged.

First, some schools in the sample have very few observations; the smallest school contains only a single student, and 25% of schools have fewer than 18 students. The random effects for these schools will be estimated with considerable uncertainty, and the multilevel model may not reliably separate within-school from between-school variation for such small clusters.

Second, PISA is a cross-sectional study in which all variables are measured at a single point in time. The associations (such as the one between `stubeha` and `zmath`) cannot be interpreted as causal. We require longitudinal data, tracking students over time to establish the temporal ordering of these relationships.

Third, the model is subject to omitted variable bias. Level 2 means of Level 1 variables were not included, and only two school-level predictors were available, leaving important characteristics such as per-pupil expenditure unaccounted for. The absence of a province variable is a notable omission given Canada's decentralised education system, as the positive effect of French language may partly reflect Quebec's strong mathematics curriculum rather than a true language effect.

There are also limitations relating to the measurement of key variables. `hisced` was treated as continuous, which assumes equal gaps between adjacent education levels. `homepos` is a composite index derived from students' self-reported household items, which is subject to biases. `immig` does not capture important details such as age at immigration or years spent in Canada.

1.6 Non-Technical Summary

Data from the study shows that around 17% of the variation in maths scores is attributable to which school a student attends, confirming that school membership has a meaningful influence on mathematical achievement over and above individual characteristics.

Family background emerges as the strongest driver of individual differences. Students from wealthier households, measured by a home possessions index, score substantially higher in mathematics. Attending a private school is associated with considerably higher maths scores even after accounting for family wealth, suggesting that school resources and selectivity play an independent role. Schools where student behaviour more negatively impacts learning tend to have lower average maths scores.

On average, girls score around 0.134 standard deviations lower than boys in mathematics. Parental education appears to benefit girls more than boys, possibly because more educated parents hold less stereotypical views about gender and invest more equally in their children's education.

Second generation immigrants score noticeably higher than native students, and first generation immigrants score similarly to or slightly above natives. This likely reflects Canada's selective immigration system, which tends to attract highly educated families with strong academic aspirations.

French-speaking students score notably higher in mathematics than English speakers. This is likely driven by Quebec's strong French-language school system, which consistently outperforms the Canadian average.

The gender gap in mathematics varies across schools; while girls generally underperform boys, some schools show the reverse. School context matters more for boys than girls, and the benefit of highly educated parents tends to be greatest in already high-performing schools.

2.1 Introduction

This study uses BHPS data to address three research questions: (i) how attitudes towards family and gender roles change with age; (ii) the extent to which individuals vary in their attitude trajectories; and (iii) what predicts these attitudes. The dataset contains 1,922 observations from 399 individuals born around 1971, followed biennially across up to nine waves (1991–2007). The response variable `zattfam` is a standardised attitude score, where higher values indicate more liberal attitudes. The two-level structure has repeated occasions at Level 1 nested within individuals at Level 2. The variables are described in Table 20. Data transformations are described in Appendix.

Since **age**, **wave**, and **year** all measure each wave of the study, **age** is used as the time variable as it's the easiest to interpret.

2.2 Descriptive Analysis

Univariate Statistics

Table 4 presents summary statistics for the continuous variables. **zattfam** has a mean of approximately 0 and standard deviation of 1, consistent with standardisation. Age ranges from 20 to 36, with a mean of 29.1 and standard deviation of 4.8.

Table 4: Summary Statistics for Continuous Variables						
Variable	Mean	SD	Min	Q1	Median	Max
zattfam	-0.000	1.000	-3.870	-0.630	0.070	3.120
age	29.13	4.79	20.00	26.00	30.00	36.00

Table 21 (Appendix) presents frequencies for the categorical variables. The majority of observations (67.9%) occur when the individual has a coresident partner. The sample is approximately balanced by gender (52.4% female). The majority of individuals hold qualifications A-level or below (81.0%), and England accounts for the largest share of individuals by country (51.4%). The number of observations per individual ranges from 2 to 9, with a median of 4 showing participant dropout.

Between- and Within-Individual Variation

To quantify the extent of between-individual variation in **zattfam**, a null variance components model was fitted with individual (**pid**) as the only random effect and no predictors:

$$\mathbf{zattfam}_{ij} = \beta_0 + u_j + e_{ij} \quad (2)$$

where $u_j \sim N(0, \sigma_u^2)$ is the individual-level random effect and $e_{ij} \sim N(0, \sigma_e^2)$ is the occasion-level residual. The estimated between-individual variance is $\hat{\sigma}_u^2 = 0.565$ and the within-individual variance is $\hat{\sigma}_e^2 = 0.436$. The variance partition coefficient (VPC), calculated as $\hat{\sigma}_u^2 / (\hat{\sigma}_u^2 + \hat{\sigma}_e^2)$, is 0.564, indicating that 56.4% of the total variation in attitude scores is attributable to stable differences between individuals. This is a high VPC, suggesting that attitudes to family are largely a persistent characteristic of the individual. The remaining 43.6% reflects within-individual change over time.

Figure 9 shows the observed **zattfam** trajectories for nine individuals. The plots show that some individuals' scores are broadly stable across waves, while others show increasing

or decreasing trends, motivating a random slope for age.

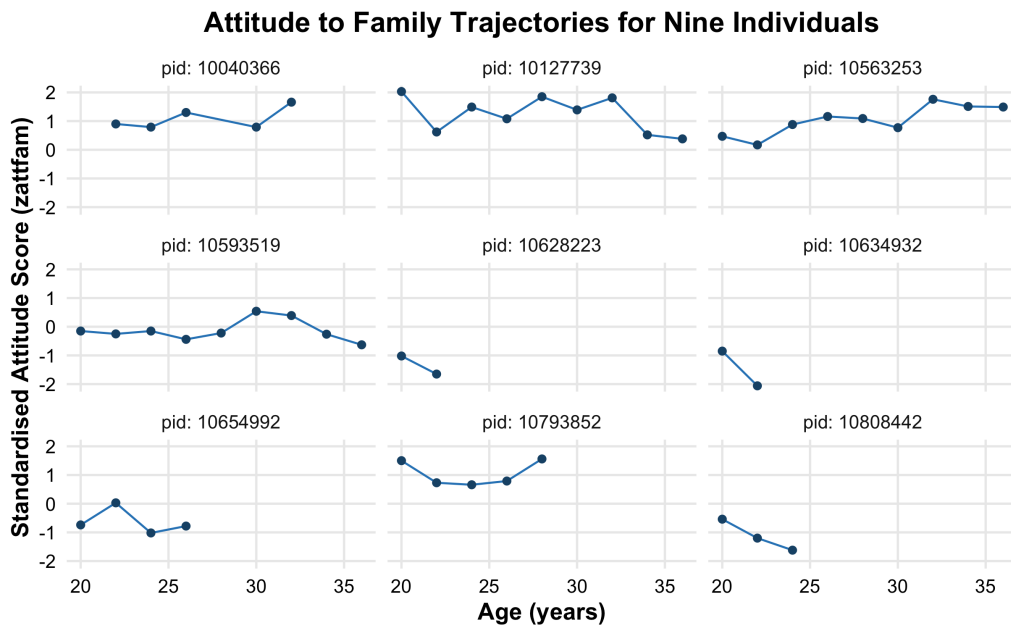


Figure 9: zattfam trajectories

Bivariate Associations

Time varying predictors

age: Figure 18 (Appendix) shows a weak negative association between age and **zattfam**, suggesting a slight tendency for attitudes to become less liberal as individuals age through their 20s and 30s.

Partnership status: Figure 10 shows that individuals with a partner have a slightly lower median attitude score than those without.

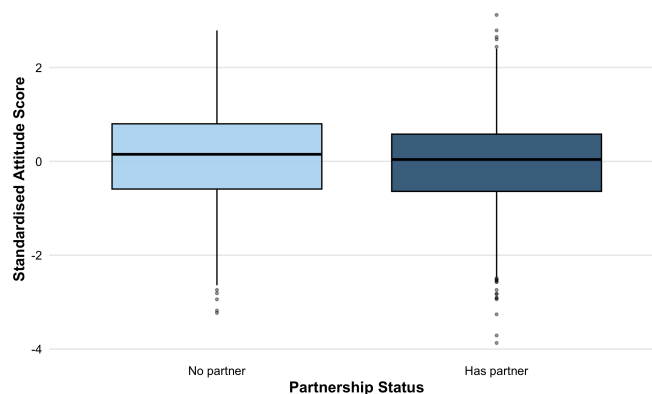


Figure 10: zattfam by partner

Time invariant predictors

For the time-invariant predictors, bivariate associations are examined using individual-level mean attitude scores to avoid conflating within- and between-individual variation.

Gender: Figure 11 shows a clear difference between males and females. Females have a noticeably higher median mean attitude score than males.

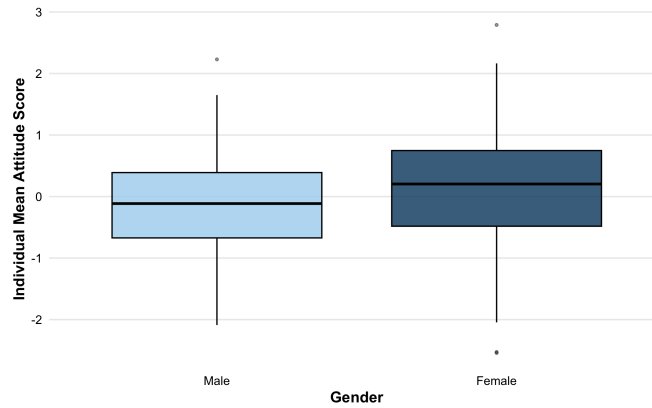


Figure 11: zattfam by female

Highest qualification: Figure 12 shows that individuals with post-school qualifications have a slightly higher median mean attitude score than those with A-level or lower.

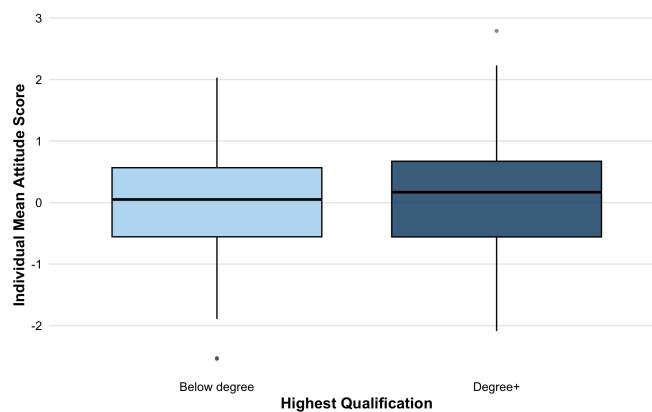


Figure 12: zattfam by qualhi

Country: Figure 13 shows little difference in median attitude scores across the four UK countries, though Northern Ireland and Scotland have wider interquartile ranges.

2.3 Model Selection

The model selection procedure follows the growth curve modelling strategy. Predictors were added sequentially and retained only if they significantly improved the model at the 5% significance level, assessed via likelihood ratio tests (LRTs).

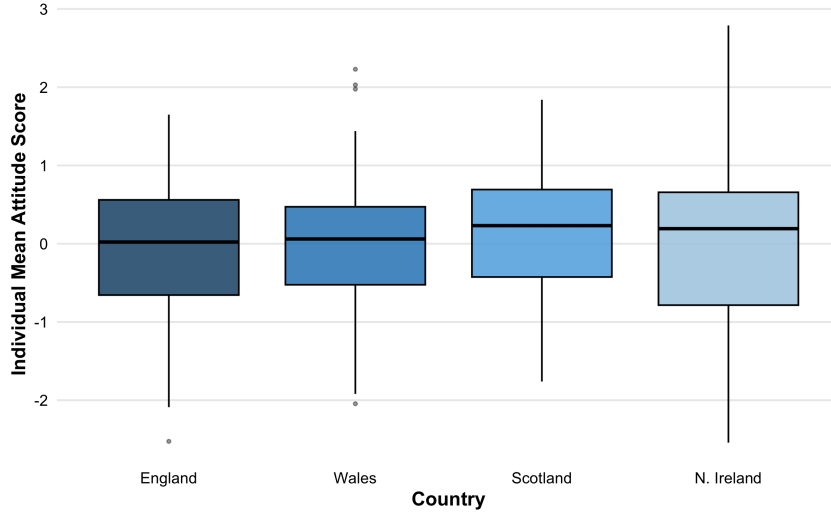


Figure 13: zattfam by country

Step 1: Null Model

The variance components model fitted in Section 2.2 serves as the baseline.

Step 2: Time Variable and Random Slope for Age

Age is added as a fixed effect and a random slope for age is tested. Since the random slope variance σ_{u1}^2 is constrained to be non-negative, a mixture chi-squared test is used, comparing the LRT statistic against $0.5\chi_1^2 + 0.5\chi_2^2$:

$$H_0 : \sigma_{u1}^2 = 0$$

$$H_1 : \sigma_{u1}^2 > 0$$

The LRT statistic is 99.336 ($p < 0.001$), so the random slope for age is retained. A quadratic age term (`age.c2`) was also tested against the random slope model via a standard LRT; it was not significant ($\chi^2 = 0.589$, $df = 1$, $p = 0.443$) and was dropped. The average trajectory is therefore assumed to be linear.

Table 5: LRT Results: Step 2

Model	LR Statistic	<i>p</i> -value
Random intercept + age.c (fixed)	—	—
+ Random slope for age.c	99.336	< 0.001***
+ age.c ² (fixed)	0.589	0.443

*** significant at 0.1%

Step 3: Time-Varying Covariate

The only level 1 covariate, **partner**, was tested against Model 2 via a standard LRT:

$$H_0 : \beta_{\text{partner}} = 0$$

$$H_1 : \beta_{\text{partner}} \neq 0$$

Table 6: LRT Results: Step 3

Predictor	LR Statistic	<i>p</i> -value
partner	4.007	0.045*

* significant at 5%

partner is significant at the 5% level and is retained in Model 3.

Step 4: Time-Invariant Covariates

The three Level 2 covariates were each tested individually against Model 3:

Table 7: LRT Results: Step 4 Individual Tests

Model	Predictor Added	LR Statistic	df	<i>p</i> -value
4.1	female	14.338	1	< 0.001***
4.2	qualhi	4.007	1	0.045*
4.3	country_lbl	5.301	3	0.151

*** significant at 0.1%; * significant at 5%

`country_lbl` is not significant and is dropped. Both `female` and `qualhi` are individually significant, so a combined model including both is fitted:

Table 8: LRT Results: Step 4 Joint Tests

Comparison	LR Statistic	df	p-value
Combined model vs Model 4.2	13.756	1	< 0.001***
Combined model vs Model 4.1	3.426	1	0.064

*** significant at 0.1%

Once both predictors are included, `female` remains highly significant, while `qualhi` falls just above the 5% threshold ($p = 0.064$), suggesting the two variables share explanatory power. `qualhi` is dropped and `female` is retained as the only time-invariant predictor.

Final Model

The final model (Model 4) includes a random intercept, a random slope for age, and fixed effects for age, partnership status, and gender:

$$\begin{aligned} \text{zattfam}_{ij} = & \beta_0 + \beta_1(\text{age.c})_{ij} + \beta_2(\text{partner})_{ij} + \beta_3(\text{female})_j \\ & + (u_{0j} + u_{1j} \text{age.c}_{ij}) + e_{ij} \end{aligned} \quad (4)$$

2.4 Model Interpretation

Fixed Effects

Table 9 presents the fixed effects estimates for the final model (Model 4). Since `zattfam` is standardised, all estimates are interpreted in standard deviation units, with each interpretation given holding all other variables constant. Reference levels are: no partner, male.

Table 9: Fixed Effects: Final Model

Variable	Estimate	Std. Error	t-value	p-value
Intercept	-0.080	0.067	-1.191	0.234
age.c	-0.012	0.005	-2.256	0.024
partner	-0.110	0.049	-2.228	0.026
female	0.310	0.081	3.835	< 0.001

Intercept: The intercept ($\hat{\beta}_0 = -0.080$, $p = 0.234$) represents the expected `zattfam` for a male, with no partner at mean age (29). It is not statistically significant, reflecting that the baseline attitude score is close to zero as expected given standardisation.

Age: The coefficient on `age.c` ($\hat{\beta}_1 = -0.012$, $p = 0.024$) indicates that each additional year of age is associated with a 0.012 standard deviation decrease in attitude score, holding other variables constant. this suggests a slight tendency for attitudes to become less liberal as individuals age. While statistically significant, the effect is small in practical terms.

Partnership status: Having a coresident partner is associated with a 0.110 standard deviation decrease in attitude score relative to those without a partner ($\hat{\beta}_2 = -0.110$, $p = 0.026$).

Gender: Females score 0.310 standard deviations higher than males on average ($\hat{\beta}_3 = 0.310$, $p < 0.001$), indicating that women hold considerably more liberal attitudes than men. This is the largest and most significant fixed effect in the model and may reflect that women are more negatively affected by gender inequalities and therefore more likely to hold more liberal views.

Random Effects

The random effects estimates are presented in Table 10.

Table 10: Random Effects: Final Model (Model 4)

Effect	Component	Variance	Std. Dev. / Correlation
Between-individual intercept	σ_{u0}^2	0.527	0.726
Age slope	σ_{u1}^2	0.004	0.061
Corr(intercept, age slope)	ρ_{u01}		0.210
Within-individual (residual)	σ_e^2	0.360	0.600

Random intercept variance: $\hat{\sigma}_{u0}^2 = 0.527$ describes the between-individual variance in attitude scores at mean age for a male with no partner. The 95% plausible range for an individual's intercept is:

$$\hat{\beta}_0 \pm 1.96\sqrt{\hat{\sigma}_{u0}^2} = -0.080 \pm 1.96 \times 0.726 = (-1.503, 1.344)$$

This wide interval reflects substantial heterogeneity in baseline attitudes between individuals.

Random slope variance: $\hat{\sigma}_{u1}^2 = 0.004$ represents the between-individual variance in the rate of attitude change with age. The 95% plausible range for the age slope is:

$$\hat{\beta}_1 \pm 1.96\sqrt{\hat{\sigma}_{u1}^2} = -0.012 \pm 1.96 \times 0.061 = (-0.132, 0.108)$$

This range spans both negative and positive values, indicating that while attitudes become slightly less liberal with age on average, some individuals' attitudes become more liberal over time. There is therefore meaningful between-individual variation in how attitudes evolve across this life stage.

Intercept-slope correlation: The correlation between the random intercept and random slope is $\hat{\rho}_{u01} = 0.210$, a weak positive correlation. This suggests that individuals who start with more liberal attitudes at mean age tend to change at a slightly faster rate, though this relationship is weak.

Between-individual variance: Since age has a random slope, the between-individual variance depends on age and is given by:

$$\hat{\sigma}_{u0}^2 + 2\hat{\sigma}_{u01}(\text{age} \cdot c) + \hat{\sigma}_{u1}^2(\text{age} \cdot c)^2$$

Figure 14 shows this as a function of age. The curve has a minimum of 0.504 at approximately age 27, rising to 0.670 at age 20 and 0.833 at age 36.

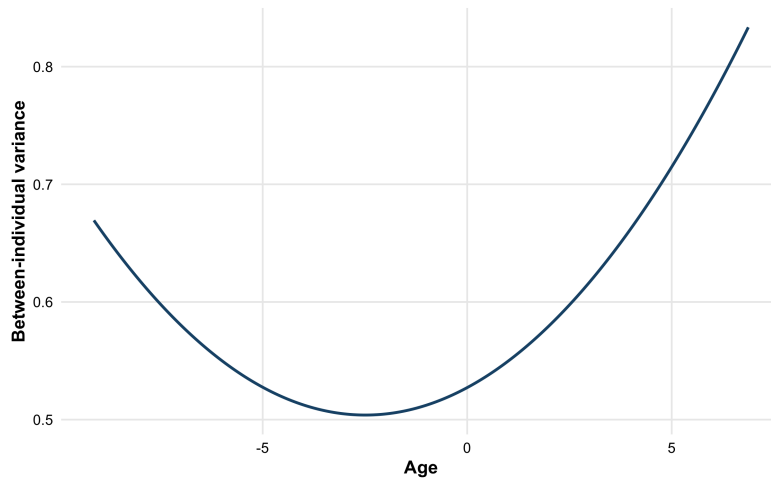


Figure 14: Between-individual variance as a function of age

2.5 Limitations

First, the number of observations per individual ranges from 2 to 9, with a median of only 4, meaning many individuals were not observed for the full study period. This may introduce bias.

Second, only one time-varying predictor (**partner**) was available at Level 1, leaving important potential predictors such as employment status and income.

Lastly, **qualhi** was treated as time-invariant despite varying for 35 of the 399 individuals across waves, hence not fitting the Level 2 structure assumed by the model.

2.6 Non-Technical Summary

This data from the study shows that over half of the variation in attitude scores (56.4%) is due to stable differences between individuals rather than changes within individuals over time. This means that people's attitudes towards family are largely a persistent personal characteristic.

Attitudes tend to become slightly less liberal as individuals age through their 20s and 30s, though this effect is small. Having a coresident partner is associated with marginally less liberal attitudes. The most notable finding is the gender difference where women hold more liberal attitudes by around 0.31 standard deviations on average compared to men. This is the largest effect in the model.

Finally, people's attitudes are most similar to each other in their mid-20s, with higher variance in early 20s and through their 30s.

Question 3

With repeated measures (level 1) nested within individuals (level 2), who are nested within countries (level 3), a three-level multilevel model is appropriate, as illustrated in Figure 15.

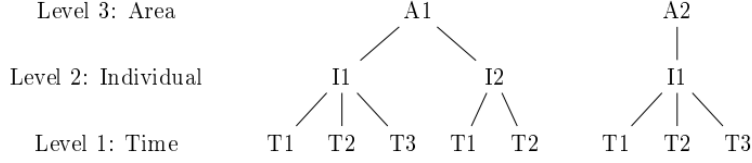


Figure 15: Three-level hierarchical structure

To investigate how individual change in attitudes with age varies across countries, the following model can be fitted:

$$\text{zattfam}_{ijk} = \beta_0 + \beta_1 \text{age}_{ijk} + u_{0jk} + u_{1jk} \text{age}_{ijk} + v_{0k} + v_{1k} \text{age}_{ijk} + e_{ijk}$$

where subscripts i, j, k denote occasion, individual, and country respectively. The random effects at each level are assumed mutually independent and normally distributed:

$$\begin{pmatrix} u_{0jk} \\ u_{1jk} \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{u0}^2 & \\ \sigma_{u01} & \sigma_{u1}^2 \end{pmatrix} \right), \quad \begin{pmatrix} v_{0k} \\ v_{1k} \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{v0}^2 & \\ \sigma_{v01} & \sigma_{v1}^2 \end{pmatrix} \right), \quad e_{ijk} \sim N(0, \sigma_e^2)$$

- β_0 : overall mean attitude at age 20, across all individuals and countries
- β_1 : average rate of change in attitudes per year of age (the fixed growth rate)
- u_{0jk} : individual-level random intercept
- u_{1jk} : individual-level random slope for age
- v_{0k} : country-level random intercept
- v_{1k} : country-level random slope for age
- σ_{u01} : covariance between individual-level intercepts and slopes
- σ_{v01} : covariance between country-level intercepts and slopes
- e_{ijk} : level 1 residual

An example research question could be to investigate whether the effect of gender on attitudes differs across EU countries. There is evidence to suggest substantial variation in **zattfam** within the EU. For example, the gender employment gap ranges from just 0.7 percentage points in Finland to 19.4 percentage points in Italy (Eurostat, 2024). A

random slope for **female** would formally test whether the gender gap in attitudes towards women and family varies significantly across EU.

Appendix

References

OECD (2024), PISA 2022 Technical Report, PISA, OECD Publishing, Paris, <https://doi.org/10.1787/01820d6d-en>.

Kao, G. and Tienda, M. (1995). Optimism and Achievement: The Educational Performance of Immigrant Youth. *Social Science Quarterly*, 76(1), 1–19.

CMEC (2022). *Measuring Up: Canadian Results of the OECD PISA 2022 Study*. Council of Ministers of Education, Canada. https://www.cmec.ca/Publications/Lists/Publications/Attachments/457/PISA2022-Highlights_EN.pdf

OECD (2024). *PISA 2022 Results (Volume I): The State of Learning and Equity in Education*. OECD Publishing, Paris. <https://doi.org/10.1787/53f23881-en>

Eurostat (2024) *Women at work: a snapshot of EU's gender employment gap*. Available at: <https://ec.europa.eu/eurostat/en/web/products-eurostat-news/w/edn-20260303-1>

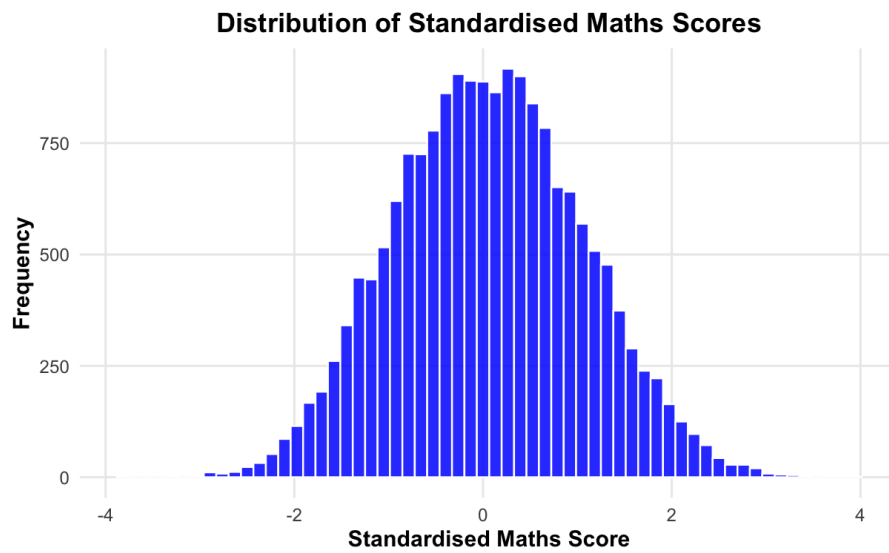


Figure 16: Distribution of Maths Scores

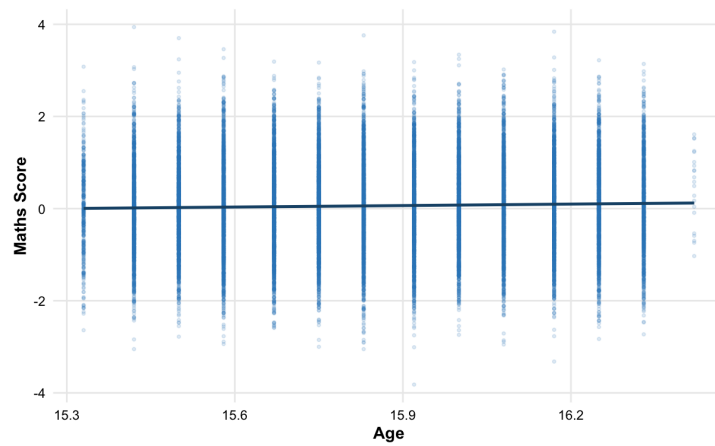


Figure 17: Maths score vs age

Table 11: Variable Descriptions

Variable	Description	Type	Ranges/Levels
<i>Individual-Level Variables</i>			
schoolid		Categorical	—
zmath	Standardised maths score	Continuous	Mean = 0, SD = 1
age	Years	Continuous	[15.33 , 16.42]
female	Gender	Binary	0 = Male* 1 = Female
immig	Immigration background	Categorical	1 = Native* 2 = 2nd Generation 3 = 1st Generation
langn	Language spoken at home	Categorical	1 = English* 2 = French 3 = Other
hisced	Highest parental education	Continuous	[2 , 10]
homepos	Home possessions score	Continuous	[-6.86 , 5.63]
<i>School-Level Variables</i>			
schprivate	School ownership type	Binary	0 = Government* 1 = Private
stubeha	Student behaviour hindering learning	Continuous	Higher = more negative impact

* indicates reference level

Table 12: Summary Statistics for Continuous Variables

Variable	Mean	SD	Min	Q1	Median	Max
<i>Individual-Level Variables</i>						
zmath	0.061	0.996	-3.820	-0.650	0.050	3.940
age	15.851	0.283	15.330	15.580	15.830	16.420
hisced	7.859	1.650	2.000	7.000	8.000	10.000
homepos	0.373	0.844	-6.860	-0.180	0.390	5.630
<i>School-Level Variables</i>						
stubeha	0.566	0.878	-3.360	0.100	0.740	3.050

Table 13: Frequencies of Categorical Variables

Variable	Category	Frequency	Percentage
<i>Individual-Level Variables</i>			
female	Male	8,906	49.5%
	Female	9,077	50.5%
immig	Native	13,308	74.0%
	2nd Generation	2,235	12.4%
	1st Generation	2,440	13.6%
langn	English	12,431	69.1%
	French	3,038	16.9%
	Other	2,514	14.0%
<i>School-Level Variables</i>			
schprivate	Government	16,602	92.3%
	Private	1,381	7.7%

Table 14: Fixed Effects: Model 2 (All Individual-Level Predictors)

Variable	Estimate	Std. Error	t-value
Intercept	−0.030	0.018	−1.65
Female	−0.136	0.013	−10.11
2nd Generation	0.110	0.025	4.43
1st Generation	0.058	0.027	2.17
French	0.127	0.026	4.80
Other language	0.058	0.026	2.22
hisced.c	0.020	0.004	4.70
homepos	0.246	0.009	28.44
age.c	0.104	0.0239	4.37

Table 15: LRT Results for Female Interactions (vs. Model 2)

Interaction with female	LR Statistic	df	<i>p</i> -value
female $\times$ immig	2.533	2	0.282
female $\times$ langn	2.253	2	0.324
female $\times$ hisced.c	8.282	1	0.004**
female $\times$ homepos	3.332	1	0.068
female $\times$ age.c	0.043	1	0.837

** significant at 1%

Table 16: LRT Results for Individual Random Slopes (vs. Model 4)

Random Slope	LR Statistic	<i>p</i> -value	Converged?
female	28.134	< 0.001***	Yes
immig	27.840	< 0.001***	No
langn	17.599	< 0.001***	No
hisced.c	41.112	< 0.001***	Yes
homepos	21.026	< 0.001***	Yes
age.c	5.501	0.042*	Yes

*** significant at 0.1%; * significant at 5%

Table 17: LRT Results for Cross-Level Interactions (vs. Model 5)

Cross-Level Interaction	LR Statistic	df	<i>p</i> -value	Converged?
hisced.c $\times$ schprivate	—	—	—	No
hisced.c $\times$ stubeha	1.225	1	0.269	Yes
female $\times$ schprivate	1.230	1	0.267	Yes
female $\times$ stubeha	1.750	1	0.186	Yes

Table 18: Fixed Effects: Final Model (Model 5)

Variable	Estimate	Std. Error	<i>t</i> -value	<i>p</i> -value
Intercept	0.003	0.022	0.147	0.883
Female	−0.134	0.014	−9.560	< 0.001
2nd Generation	0.107	0.025	4.362	< 0.001
1st Generation	0.051	0.026	1.944	0.052
French	0.129	0.026	5.016	< 0.001
Other language	0.057	0.026	2.165	0.030
hisced.c	0.009	0.006	1.418	0.156
homepos	0.242	0.009	28.068	< 0.001
age.c	0.108	0.024	4.551	< 0.001
Private school	0.345	0.060	5.721	< 0.001
stubeha	−0.098	0.017	−5.832	< 0.001
Female × hisced.c	0.027	0.008	3.274	0.001

Table 19: Random Effects Estimates: Final Model

Effect	Component	Variance	Std. Dev. / Correlation
Between-school intercept	σ_{u0}^2	0.1287	0.3588
Female slope	σ_{u1}^2	0.0117	0.1080
hisced.c slope	σ_{u2}^2	0.0030	0.0548
Corr(intercept, female)	ρ_{u01}		−0.787
Corr(intercept, hisced.c)	ρ_{u02}		0.504
Corr(female, hisced.c)	ρ_{u12}		−0.739
Within-school (residual)	σ_e^2	0.7716	0.8784

Model 4

$$\begin{aligned}
\text{zmath}_{ij} = & \beta_0 + \beta_1(\text{Female})_{ij} + \beta_2(\text{2nd Gen})_{ij} + \beta_3(\text{1st Gen})_{ij} \\
& + \beta_4(\text{French})_{ij} + \beta_5(\text{Other})_{ij} + \beta_6(\text{hiscd.c})_{ij} \\
& + \beta_7(\text{homepos})_{ij} + \beta_8(\text{age.c})_{ij} + \beta_9(\text{Female} \times \text{hiscd.c})_{ij} \\
& + \beta_{10}(\text{schprivate})_j + \beta_{11}(\text{stubeha})_j + u_j + e_{ij}
\end{aligned} \tag{4}$$

Table 20: Variable Descriptions

Variable	Description	Codes
<i>Time-Varying Variables (Level 1)</i>		
zattfam	Standardised attitude to family score	Continuous
age	Years	Continuous
partner	Has a coresident partner	0 = No*, 1 = Yes
<i>Time-Invariant Variables (Level 2)</i>		
female	Gender	0 = Male*, 1 = Female
qualhi	Highest qualification	0 = A-level or lower*, 1 = Post-school
country	Country of residence	1 = England*, 2 = Wales, 3 = Scotland, 4 = N. Ireland

* indicates reference level

Table 21: Frequencies of Categorical Variables

Variable	Category	Frequency	Percentage
<i>Level 1 % of observations</i>			
partner	No partner	617	32.1%
	Has partner	1305	67.9%
<i>Level 2 % of individuals</i>			
female	Male	190	47.6%
	Female	209	52.4%
qualhi	A-level or lower	323	81.0%
	Post school	76	19.0%
country	England	205	51.4%
	Wales	56	14.0%
	Scotland	86	21.6%
	N. Ireland	52	13.0%

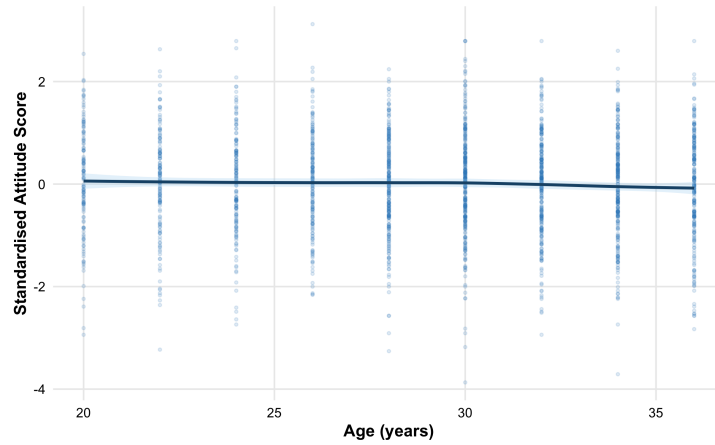


Figure 18: Attitude score vs age

Data Transformations

Question 1

Two continuous predictors were transformed prior to model fitting. `age` and `hisced` were mean centred (`age.c = age - mean(age)`) and (`hisced.c = hisced - mean(hisced)`). Centring ensures the intercept is interpreted at a meaningful value as the estimated outcome for a student of average age and average parental education. `stubeha` was standardised (`stubeha.z = scale(stubeha)`) to address convergence issues encountered when fitting random slope models.

Question 2

Age was mean centred (`age.c = age - mean(age)`), so β_0 reflects the estimated mean attitude score at the average age in the sample (approximately 29.1 years) rather than at an unobserved value of zero. A squared term (`age.c2 = age.c2`) was used to test non-linearity.